

Supplementary Material

Well-posedness, Stability, and Bifurcation in an Acid-Mediated Tumour
Invasion Model
with Interspecific Competition

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This document contains the technical details excluded from the main manuscript for brevity: the full derivation of the nondimensional system, the coefficient-by-coefficient expansion behind the spatial-stability proposition, the extended versions of the remarks on the gauge and on degeneracy, the extended unfolding of the codimension-two point, the Volterra functional on branch (i) with its optimal weights, the properties of the threshold set of the spatial problem, the numerical controls behind the size threshold, the Statements and Declarations section, and the symbolic verifications performed with SymPy. All proofs are now given in the main text; see Section 2. References to theorems, lemmas, propositions, equations and remarks of the main manuscript are given in the form “Theorem 3.X”, “Proposition 3.Y”, etc.; references of the form “Theorem S.X” and equations of the form “(S.k)” refer to the present supplementary material.

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1 Full derivation of the nondimensional system

Starting from the dimensional system

$$\begin{aligned}\partial_t \tilde{N} &= \tilde{r}_1 \tilde{N} \left(1 - \tilde{N}/\tilde{K}_N\right) - \tilde{d}_1 \tilde{H} \tilde{N} + \tilde{D}_1 \nabla_{\tilde{x}}^2 \tilde{N}, \\ \partial_t \tilde{T} &= \tilde{r}_2 \tilde{T} \left(1 - \tilde{T}/\tilde{K}_T - \tilde{\alpha} \tilde{N}/\tilde{K}_N\right) + \tilde{D}_2 \nabla_{\tilde{x}}^2 \tilde{T}, \\ \partial_t \tilde{H} &= \tilde{r}_3 \tilde{T} - \tilde{d}_3 \tilde{H} + \tilde{D}_3 \nabla_{\tilde{x}}^2 \tilde{H},\end{aligned}\tag{S.1}$$

we introduce the rescaled variables

$$N = \tilde{N}/\tilde{K}_N, \quad T = \tilde{T}/\tilde{K}_T, \quad H = \tilde{H}/\tilde{H}_0, \quad t = \tilde{r}_1 \tilde{t}, \quad x = \tilde{x}/\tilde{X}_0, \quad \tilde{X}_0 = \sqrt{\tilde{D}_3/\tilde{r}_1}.$$

The reference acid concentration $\tilde{H}_0 > 0$ is kept free at this stage. Dividing the first equation of (S.1) by $\tilde{r}_1 \tilde{K}_N$ and using $\nabla_{\tilde{x}}^2 = \tilde{X}_0^{-2} \nabla^2$, one obtains $\partial_t N = N(1 - N) - (\tilde{d}_1 \tilde{H}_0/\tilde{r}_1) H N + (\tilde{D}_1/\tilde{D}_3) \nabla^2 N$. Setting $\delta = \tilde{d}_1 \tilde{H}_0/\tilde{r}_1$ and $\delta_1 = \tilde{D}_1/\tilde{D}_3$ yields the first equation of the nondimensional system. The second equation becomes, after division by $\tilde{r}_1 \tilde{K}_T$, $\partial_t T = (\tilde{r}_2/\tilde{r}_1) T(1 - T - \alpha N) + (\tilde{D}_2/\tilde{D}_3) \nabla^2 T$ with $\beta = \tilde{r}_2/\tilde{r}_1$ and $\delta_2 = \tilde{D}_2/\tilde{D}_3$. The third equation, divided by $\tilde{r}_1 \tilde{H}_0$, takes the form $\partial_t H = (\tilde{r}_3 \tilde{K}_T/(\tilde{r}_1 \tilde{H}_0)) T - (\tilde{d}_3/\tilde{r}_1) H + \nabla^2 H$, with $\gamma = \tilde{r}_3 \tilde{K}_T/(\tilde{r}_1 \tilde{H}_0)$ and $\omega = \tilde{d}_3/\tilde{r}_1$. The critical threshold $\delta_c = \omega/\gamma = \tilde{d}_3/(\tilde{r}_3 \tilde{K}_T/\tilde{H}_0)$ is the ratio of acid clearance to acid supply per unit reference concentration. It is *not* itself invariant under the rescaling $\tilde{H}_0 \mapsto \lambda \tilde{H}_0$; what is invariant is the position of δ relative to it, i.e. the ratio δ/δ_c , as discussed in Remark S1.1 below.

Remark S1.1 (Gauge invariance; extended version of Remark 2.1 of the manuscript). A change $\tilde{H}_0 \mapsto \lambda \tilde{H}_0$ with $\lambda > 0$ rescales $H \mapsto H/\lambda$ and acts on the nondimensional parameters as $\gamma \mapsto \gamma/\lambda$, $\delta \mapsto \lambda \delta$, while ω is unchanged. The form of the system is preserved, but the individual parameters are not: δ and $\delta_c = \omega/\gamma$ both carry a factor λ , since $\delta_c \mapsto \lambda \delta_c$. What survives the rescaling is the product $\delta\gamma$ and, equivalently, the ratio $\delta/\delta_c = \delta\gamma/\omega$, which is the only combination of (γ, δ, ω) our results depend on. In particular, saying that a parameter set lies below or above threshold is a gauge-independent statement even though neither δ nor δ_c is a gauge-independent number. The natural choice $\tilde{H}_0 = \tilde{r}_3 \tilde{K}_T/\tilde{d}_3$ (acid concentration produced at saturation, balanced by clearance) yields $\gamma = \omega$ and $\delta_c = 1$, reducing the parameter count by one and recovering the single-parameter bifurcation of the original Gatenby–Gawlinski model. We keep $\tilde{H}_0$ free for two reasons. First, it makes the dimensional roles of acid production and acid clearance explicit, which simplifies the comparison of tumour phenotypes with distinct glycolytic activity $\tilde{r}_3$ and distinct vascular buffering $\tilde{d}_3$. Second, all our analytical statements are phrased in terms of gauge-invariant quantities, so the choice of $\tilde{H}_0$ has no bearing on the qualitative results: the two-parameter pair (γ, ω) is a notational convenience, not an additional mechanism.

2 Proofs now given in the main text

In the present version, the proofs of the invariance theorem, of the L^∞ lemma for weak solutions, of the global existence and uniqueness theorem, of the asymptotic H^1 regularity lemma and of the two global stability theorems are given *in full in the main text*, and are no longer reproduced here. They were placed in this Supplementary Material in the first submitted version; both referees reported that the Supplementary Material had not

been made available to them, and were accordingly unable to assess arguments that the manuscript presented only as sketches. We have therefore moved the complete arguments back into the body of the paper, where they belong, and this document is reduced to the material that genuinely supplements it: the nondimensionalisation, the extended discussion of the codim-2 point, the Volterra functional, the threshold set of the spatial problem, the symbolic verifications, and the declarations.

Two of the proofs became shorter in the process. The exponential decay of the normal-cell density in the proof of the global stability of E_2 is now obtained directly in L^∞ , by comparison with a spatially homogeneous supersolution, instead of passing through an L^1 estimate and a parabolic L^p – L^q smoothing argument; and the lower bound $\liminf T \geq 1 - \alpha$ now follows from an elementary comparison with a logistic ordinary differential equation, without invoking travelling-front theory on bounded domains.

3 Coefficient expansion for the spatial-stability proposition

The proof is now given in full in the main text; we record here the cofactor computation on which it rests, together with the degree-by-degree expansion that it replaces.

Let E_3 lie on branch (i), write $J_k = J(E_3) - \mu D$ with $D = \text{diag}(\delta_1, \delta_2, 1)$ and $\mu = \mu_k \geq 0$ an eigenvalue of $-\Delta_N$, and set

$$A = N^* + \delta_1 \mu, \quad B = \beta T^* + \delta_2 \mu, \quad C = \omega + \mu, \quad \kappa = \alpha \beta \delta \gamma N^* T^* > 0.$$

Since J_k differs from $J(E_3)$ only in its diagonal, its off-diagonal entries are those of (3.14): the three products $(J_k)_{ij}(J_k)_{ji}$ vanish identically, and the unique non-trivial directed cycle of the interaction graph is $N \rightarrow T \rightarrow H \rightarrow N$, of weight $(-\alpha \beta T^*) \cdot \gamma \cdot (-\delta N^*) = \kappa$. Cofactor expansion of $\det(J_k - \lambda I)$ therefore gives $\lambda^3 + \tilde{a}_1 \lambda^2 + \tilde{a}_2 \lambda + \tilde{a}_3$ with

$$\tilde{a}_1 = A + B + C, \quad \tilde{a}_2 = AB + BC + CA, \quad \tilde{a}_3 = ABC - \kappa,$$

the middle coefficient receiving no cross term precisely because there is no 2-cycle. Expanding in powers of μ recovers the coefficient-by-coefficient form

$$\tilde{a}_1(\mu) = a_1 + \mu(\delta_1 + \delta_2 + 1), \tag{S.2}$$

$$\tilde{a}_2(\mu) = a_2 + \mu[N^*(\delta_2 + 1) + \beta T^*(\delta_1 + 1) + \omega(\delta_1 + \delta_2)] + \mu^2(\delta_1 \delta_2 + \delta_1 + \delta_2), \tag{S.3}$$

$$\tilde{a}_3(\mu) = a_3 + \mu[N^* \beta T^* + \omega(N^* \delta_2 + \beta T^* \delta_1)] + \mu^2[N^* \delta_2 + \beta T^* \delta_1 + \delta_1 \delta_2 \omega] + \mu^3 \delta_1 \delta_2, \tag{S.4}$$

each difference $\tilde{a}_i(\mu) - a_i$ being a polynomial with strictly positive coefficients; but the closed form above is what makes the conclusion immediate. The coefficients A, B, C increase with μ , so $\tilde{a}_3(\mu) \geq \tilde{a}_3(0) = \beta N^* T^* (\omega - \alpha \delta \gamma) > 0$ on branch (i), while the elementary identity $(A + B + C)(AB + BC + CA) - ABC = (A + B)(B + C)(C + A)$ yields

$$\tilde{a}_1 \tilde{a}_2 - \tilde{a}_3 = (A + B)(B + C)(C + A) + \kappa > 0 \quad (\mu \geq 0).$$

The Routh–Hurwitz criterion holds for every mode, so no mode can destabilise E_3 , for any $\delta_1, \delta_2 > 0$. Setting $\mu = 0$ returns the Hopf margin of the main text: the absence of Hopf bifurcation is the zero-mode instance of the absence of Turing instability. All four identities are checked symbolically in `sympy_verification.py`, Section 9.

4 Extended codim-2 unfolding and degenerate cases

Degenerate configurations (extended Remark 3.23)

On the boundary of the existence region of E_3 , the formulas (3.10) of the main manuscript produce coalescing equilibria rather than new branches. (i) When $\alpha = 1$, the explicit formulas give $T^* = H^* = 0$ and $N^* = 1$, so E_3 collapses onto E_1 ; this is the transcritical curve Γ_α , and the collapse occurs for every $\delta \neq \delta_c$, from branch (i) when $\delta < \delta_c$ and from branch (ii) when $\delta > \delta_c$. (ii) When $\delta = \delta_c$, one has $N^* = 0$, $T^* = 1$, $H^* = \gamma/\omega$, so E_3 collapses onto E_2 ; this is the transcritical curve Γ_δ , again approached from either branch according to the sign of $1 - \alpha$. (iii) When $\alpha = 1$ and $\delta = \delta_c$ simultaneously, both transcritical curves meet at the codimension-two organising centre P^* , which is also the point where the coexistence state changes dynamical type, passing from attractor (branch (i)) to saddle (branch (ii)). Here the two collapses of (i) and (ii) cannot both be read off the formulas, and for a reason: at P^* the equilibrium set is no longer finite but contains the whole segment $\mathcal{S}$ joining E_1 to E_2 , onto which E_3 collapses in a direction-dependent way. This is the content of Proposition 3.24 of the main text, and it is what the unfolding below is built on. (iv) The denominator $q = \omega - \alpha\delta\gamma$ of (3.10) vanishes on the hyperbola $\delta = \delta_c/\alpha$, which meets neither branch: it lies strictly above δ_c when $\alpha < 1$ and strictly below δ_c when $\alpha > 1$, so in both cases it falls where E_3 is inadmissible. The singularity is apparent and creates no dynamical event.

Sotomayor coefficients on Γ_α

At $E_1 = (1, 0, 0)$ for $\alpha = 1$, with α as the bifurcation parameter, the right and left null vectors of $J(E_1; \alpha = 1)$ are $v = (-\delta, \omega/\gamma, 1)^\top$ and $w = (0, 1, 0)^\top$. The three Sotomayor coefficients evaluate to

$$w^\top F_\alpha = 0, \quad w^\top (DF)_\alpha v = -\beta\omega/\gamma \neq 0, \quad w^\top D^2F(v, v) = 2\beta\omega(\delta\gamma - \omega)/\gamma^2.$$

For $\delta \neq \delta_c = \omega/\gamma$ the third coefficient is non-zero, and the bifurcation along Γ_α is a non-degenerate transcritical exchange of stability between E_1 and E_3 . At $\delta = \delta_c$ it vanishes, mirroring the degeneracy observed on Γ_δ at $\alpha = 1$, and flagging the codim-2 point P^* . Its sign, on the other hand, carries no invariant meaning: v and w are defined up to scalars, and dividing by $w^\top v$ does not repair this, since $w^\top D^2F(v, v)$ is quadratic in v where $w^\top v$ is linear, so that the quotient picks up a factor c under $v \mapsto cv$. The convention-free object is the slope of the bifurcating branch, $-2\sigma_1 v/\sigma_2$ with $\sigma_1 = w^\top (DF)_\mu v$ and $\sigma_2 = w^\top D^2F(v, v)$, in which both scalars cancel; on Γ_δ its first component returns $dN^*/d\delta = -\gamma/(\omega(1 - \alpha))$ at $\delta = \delta_c$, in agreement with the closed form, and Section 8 of the script verifies both the identity and the failure of invariance.

Unfolding at P^*

The two Sotomayor coefficients that vanish simultaneously at P^* are the *quadratic* ones, $w^\top D^2F(v, v)$, computed on each of the two transcritical curves; the transversality coefficients do not vanish there. Their simultaneous vanishing signals that P^* is not an ordinary point of either curve, but it does not by itself identify the local structure. The identification is nevertheless elementary, because the centre direction at P^* is one-dimensional: E_1 and E_2 each carry a simple zero eigenvalue there ($\sigma(J_1) = \{-1, -\omega, 0\}$).

and $\sigma(J_2) = \{-\beta, -\omega, 0\}$, and the centre manifold is the segment $\mathcal{S}$ of (3.32). The relevant tool is therefore Fenichel's theory of normally hyperbolic invariant manifolds, and the unfolding closes in a few lines. Proposition 3.24 of the main text establishes that $\mathcal{S}$ is a compact, attracting, normally hyperbolic manifold of equilibria, and the reduced scalar field (3.39) obtained by projection on its tangent bundle reproduces both admissible branches with their stability types.

Overflowing invariance by enlargement. Fenichel's theorem, and that of Hirsch, Pugh and Shub [1, 2, 3], require of a compact manifold *with boundary* that the flow cross $\partial\mathcal{S}$ transversally (overflowing or inflowing invariance), whereas the field vanishes identically on $\partial\mathcal{S} = \{E_1, E_2\}$: invariance of the boundary is weaker than transversality, and is precisely the degenerate case the hypothesis excludes. The repair is in two steps.

First, enlarge. Nothing in the proof of Proposition 3.24 used $s \in [0, 1]$: the reaction field vanishes on the whole affine line (3.33), and $A(s)$ and $B(s)$ are continuous and strictly positive on the compact $[0, 1]$, hence bounded below there by a positive constant, so by uniform continuity on $[-1, 2]$ there is $\varepsilon_0 > 0$ with $A, B > 0$ on $[-\varepsilon_0, 1 + \varepsilon_0]$. For $\varepsilon \leq \varepsilon_0$ the enlarged segment $\mathcal{S}^\varepsilon := \Lambda|_{[-\varepsilon, 1+\varepsilon]}$ is again a compact manifold of equilibria whose transverse eigenvalues satisfy $\operatorname{Re} \lambda \leq -c$ for some $c > 0$, uniformly in s .

Second, modify. Enlarging alone does not restore transversality, the field vanishing on $\partial\mathcal{S}^\varepsilon$ too, but it creates a buffer where the vector field may be modified, away from $\Lambda|_{[0,1]}$, so as to point strictly outward across $\partial\mathcal{S}^\varepsilon$. The modified system has $\mathcal{S}^\varepsilon$ as an overflowing invariant normally hyperbolic manifold, so the persistence theorem applies; and the modification being supported away from $\Lambda|_{[0,1]}$, the conclusion drawn over $[0, 1]$ is independent of it. For (a, d) near the origin there is therefore a locally invariant attracting curve $\mathcal{S}_{a,d}$, C^r for each finite r , depending smoothly on the parameters and $O(|a| + |d|)$ -close to $\mathcal{S}$. Its endpoints may be taken to be E_1 and E_2 themselves, equilibria of F for *every* (α, δ) , so $\partial\mathcal{S}_{a,d}$ is invariant after perturbation as well. Being attracting, $\mathcal{S}^\varepsilon$ carries no unstable fibres, and the maximal invariant set of the neighbourhood is $\mathcal{S}_{a,d}$ itself: an equilibrium lying near $\mathcal{S}$ remains in the neighbourhood for all *negative* times as well, hence belongs to it.

Three verifications underpin the reduction, and all three are performed symbolically in `sympy_verification.py`, Section 11. First, the reaction field vanishes identically on $\mathcal{S}$ at $(\alpha, \delta) = (1, \delta_c)$, and the Jacobian along $\mathcal{S}$ annihilates the tangent vector $\tau = (1, -1, -\gamma/\omega)^\top$, its characteristic polynomial factoring as $\lambda(\lambda^2 + A(s)\lambda + B(s))$ with $A(s) = s + \beta(1 - s) + \omega$ and $B(s) = \omega s + \beta(1 - s)(\omega + s)$, both positive on $[0, 1]$. Second, under $\alpha = 1 + a$ and $\delta = \delta_c(1 + d)$ the denominator of (3.10) becomes $q = -\omega(a + d + ad)$, whence $N^* = d/(a + d + ad)$ and $T^* = a/(a + d + ad)$. That denominator vanishes on a curve through the origin of tangent $d = -a$, along which E_3 is unbounded; along every ray of direction (a_0, d_0) with $a_0 + d_0 \neq 0$ one has $N^* + T^* \rightarrow 1$, and the limiting point $s_0 = d_0/(a_0 + d_0)$ lies on $\mathcal{S}$ exactly when $a_0 d_0 \geq 0$, that is, under the admissibility condition. Third, the perturbed field restricted to $\mathcal{S}$ is exactly $-s(1 - s)(d, a\beta, 0)^\top$, which is what makes the projection along the stable fibres elementary and produces the reduced field (3.39).

A numerical control accompanies the symbolic one, and is performed by the same script. Integrating the full three-dimensional kinetics and the reduced scalar equation from the same initial datum on $\mathcal{S}$ (taken at $s = \frac{1}{2}$), at $(a, d) = \varepsilon(a_0, d_0)$ with $\varepsilon \in \{10^{-2}, 5 \cdot 10^{-2}\}$ and $(a_0, d_0) \in \{(1, 2), (-1, -2), (1, \frac{1}{2})\}$, the two models select the same equilibrium in all six cases. On branch (ii) the interior root repels and both leave for the same endpoint; on

branch (i) both settle on the interior root, and the residual gap is exactly the displacement $s^* - N^* = ad^2/[(a+d)(a+d+ad)]$ between the root of the leading-order field and the exact coexistence state, which is $-2.30 \cdot 10^{-2}$ at $\varepsilon = 5 \cdot 10^{-2}$.

5 The Volterra functional on branch (i)

Theorem 3.17 of the main text settles convergence but says nothing about how it happens. The Volterra functional below does, on the sub-region of branch (i) where an explicit sufficient condition holds, in the form introduced for Lotka–Volterra systems by Goh [4] and made standard in mathematical biology by [5].

Hypothesis S5.1 (HL). The parameters satisfy the two conditions

$$\alpha^2\beta < 4 \quad \text{and} \quad \Delta(\alpha, \beta, \gamma, \delta, \omega) > 0, \quad (\text{S.5})$$

where

$$\Delta(\alpha, \beta, \gamma, \delta, \omega) := 4\beta\omega - \gamma^2 - \alpha^2\beta^2\omega - \alpha\beta\delta\gamma - \beta\delta^2. \quad (\text{S.6})$$

Remark S5.2. Hypothesis (HL) is the Sylvester positive-definiteness criterion of the matrix arising from the Volterra functional with canonical weights $c_1 = c_2 = c_3 = 1$. It bears on the certificate rather than on the result, since Theorem 3.17 of the main text already gives convergence to E_3 on the whole of branch (i); what (HL) buys is a monotone quantity along the flow. It is also, unlike the branch condition, not invariant under the rescaling of Remark 2.1 of the main text. For the default regime of Section 4 of the main text ($\alpha = 0.3$, $\beta = 1.5$, $\gamma = 5$, $\omega = 8$, $\delta = 0.5$), $\alpha^2\beta = 0.135$ and $\Delta \approx 19.88$, so (HL) is satisfied with room to spare.

Before the global-stability theorem we record the persistence estimate that makes the Volterra functional a *bona fide* Lyapunov functional: it keeps N and T uniformly away from zero, so that $\ln(N/N^*)$ and $\ln(T/T^*)$ stay integrable and the functional finite and continuous on the absorbing set. It is proved by exactly the comparison technique used later for E_2 , and it is here that the two defining inequalities of branch (i) do their work.

Lemma S5.3 (Uniform persistence on branch (i)). *Let $\alpha < 1$ and $\delta < \delta_c$, and let u be the solution of (2.2) with $u_0 \in \mathcal{R}$ and $N_0, T_0 \geq c_0 > 0$ a.e. Then there exist constants $\underline{N}, \underline{T} > 0$, depending only on the parameters, and a time $t_* = t_*(u_0)$ such that*

$$N(x, t) \geq \underline{N}, \quad T(x, t) \geq \underline{T}, \quad \text{for a.e. } x \in \Omega, \quad t \geq t_*. \quad (\text{S.7})$$

Explicitly one may take $\underline{T} = (1 - \alpha)/2$ and $\underline{N} = \frac{1}{2}(1 - \delta/\delta_c)$, both strictly positive exactly because $\alpha < 1$ and $\delta < \delta_c$.

Proof. Every step applies the scalar comparison Lemma 3.3 of the main text with a spatially homogeneous barrier.

Since $0 \leq N \leq 1$, the tumour equation obeys $\partial_t T \geq \beta T(1 - \alpha - T) + \delta_2 \nabla^2 T$; comparison with the logistic ODE of carrying capacity $1 - \alpha > 0$, started at $\min(c_0, 1 - \alpha)$, gives $\liminf_t \inf_x T \geq 1 - \alpha$, so $T \geq 1 - \alpha - \varepsilon \geq \underline{T}$ for t large.

For the acid only an upper bound is needed, it entering the N -equation with a minus sign. From $T \leq 1$, $\partial_t H \leq \gamma - \omega H + \nabla^2 H$, and Lemma 3.3 of the main text against the decreasing supersolution $\bar{v}(t) = \gamma/\omega + (\|H_0\|_\infty - \gamma/\omega)^+ e^{-\omega t}$ gives $\limsup_t \sup_x H \leq \gamma/\omega$, hence $H \leq \gamma/\omega + \varepsilon$ for t large.

Finally $\partial_t N \geq N(1 - \delta(\gamma/\omega + \varepsilon) - N) + \delta_1 \nabla^2 N$, whose logistic carrying capacity $1 - \delta/\delta_c - \delta\varepsilon$ is strictly positive for small ε exactly because $\delta < \delta_c$. Starting the comparison needs N bounded away from 0 at some time: since $N \geq 0$ and, on $\mathcal{R}$, $N(1 - N - \delta H) \geq -\delta M_0 N$, the constant $c_0 e^{-\delta M_0 t}$ is a subsolution (Lemma 3.3 of the main text), so $N(x, t) \geq c_0 e^{-\delta M_0 t} > 0$ for every finite t . Under the uniform lower bound assumed here, neither connectedness nor heat-kernel positivity enters, in contrast with Theorem 3.18 of the main text, which starts from $N_0 \not\equiv 0$ and needs both. Comparison with the logistic ODE above then gives $\liminf_t \inf_x N \geq 1 - \delta/\delta_c$, hence $N \geq \underline{N}$ for t large. $\square$

Theorem S5.4 (Global stability of E_3). *Assume that (α, δ) lies on branch (i) of (3.11), i.e. $\alpha < 1$ and $\delta < \delta_c$, and that Hypothesis (HL) holds. Let $u_0 \in \mathcal{R}$ with $N_0, T_0 \geq c_0 > 0$ a.e. Then $\mathcal{L}$ of (S.8) is a strict Lyapunov functional on the absorbing set, and the solution of (2.2) satisfies*

$$(N(\cdot, t), T(\cdot, t), H(\cdot, t)) \xrightarrow[t \rightarrow \infty]{L^2(\Omega)} (N^*, T^*, H^*).$$

Proof. Define the Volterra-type functional

$$\mathcal{L}(t) = \int_{\Omega} \left[\left(N - N^* - N^* \ln \frac{N}{N^*} \right) + \left(T - T^* - T^* \ln \frac{T}{T^*} \right) + \frac{1}{2} (H - H^*)^2 \right] dx, \quad (\text{S.8})$$

which is non-negative and vanishes exactly at E_3 . Its finiteness and continuity on the phase space are not automatic, the entropy densities $N - N^* - N^* \ln(N/N^*)$ blowing up as $N \downarrow 0$, and this is where Lemma S5.3 is indispensable: for $t \geq t_*$ the orbit lies in the absorbing set

$$\mathcal{R}_* = [\underline{N}, 1] \times [\underline{T}, 1] \times [0, \gamma/\omega + \varepsilon], \quad \underline{N}, \underline{T} > 0,$$

on which N, T are bounded away from 0 uniformly in time. There $u \mapsto \mathcal{L}$ is finite and Lipschitz for the L^2 topology and $\nabla(N^*/N) = -N^* \nabla N / N^2 \in L^2$, so the chain rule below is licit. We run the argument on $[t_*, \infty)$, which is all a statement about $t \rightarrow \infty$ requires.

Step 1 (the reactive part). Write $n = N - N^*$, $\tau = T - T^*$, $h = H - H^*$ and $\xi = (n, \tau, h)^\top$. Differentiating (S.8) along the flow, the reaction terms contribute

$$\int_{\Omega} \left[\frac{n}{N} N(1 - N - \delta H) + \frac{\tau}{T} \beta T(1 - T - \alpha N) + h(\gamma T - \omega H) \right] dx.$$

The equilibrium relations $1 - N^* - \delta H^* = 0$, $1 - T^* - \alpha N^* = 0$ and $\gamma T^* = \omega H^*$ rewrite each bracket in the deviations as $-n - \delta h$, $-\tau - \alpha n$ and $\gamma \tau - \omega h$, so the reactive part equals

$$\left. \frac{d\mathcal{L}}{dt} \right|_{\text{react}} = - \int_{\Omega} [n^2 + \beta \tau^2 + \omega h^2 + \delta n h + \alpha \beta \tau n - \gamma \tau h] dx = - \int_{\Omega} \xi^\top Q \xi dx, \quad (\text{S.9})$$

with the symmetric matrix

$$Q = \begin{pmatrix} 1 & \alpha\beta/2 & \delta/2 \\ \alpha\beta/2 & \beta & -\gamma/2 \\ \delta/2 & -\gamma/2 & \omega \end{pmatrix}.$$

Step 2 (the diffusive part is a gift). The diffusion terms contribute

$$\int_{\Omega} \left[\frac{n}{N} \delta_1 \nabla^2 N + \frac{\tau}{T} \delta_2 \nabla^2 T + h \nabla^2 H \right] dx = -\delta_1 N^* \int_{\Omega} \frac{|\nabla N|^2}{N^2} - \delta_2 T^* \int_{\Omega} \frac{|\nabla T|^2}{T^2} - \int_{\Omega} |\nabla H|^2 \leq 0,$$

after integrating by parts, the boundary terms vanishing by the Neumann condition, and using $\nabla(n/N) = N^* \nabla N / N^2$ and its analogue for T . Both integrals converge because $N \geq \underline{N}$ and $T \geq \underline{T}$ uniformly in time on $\mathcal{R}_*$ (Lemma S5.3), so $|\nabla N|^2 / N^2 \leq \underline{N}^{-2} |\nabla N|^2$ and likewise for T .

Step 3 (Sylvester, and where (HL) comes from). Cofactor expansion of Q gives its three leading principal minors,

$$m_1 = 1, \quad m_2 = \beta - \frac{\alpha^2 \beta^2}{4} = \frac{\beta(4 - \alpha^2 \beta)}{4}, \quad m_3 = \det Q = \frac{\Delta}{4},$$

with $\Delta = 4\beta\omega - \gamma^2 - \alpha^2\beta^2\omega - \alpha\beta\delta\gamma - \beta\delta^2$ exactly as in (S.6). Hypothesis (HL) is therefore exactly Sylvester's criterion for $Q \succ 0$, reached without any Young inequality, and at the canonical weights $c_i = 1$ it makes the reactive part a strict dissipation term. It asks more than $d\mathcal{L}/dt \leq 0$ requires, the diffusive part being strictly negative, and it is not gauge-invariant (Remark 2.1 of the main text), so other weights certify a larger region. Under (HL) the integrand of (S.9) is a positive definite quadratic form in ξ , so $d\mathcal{L}/dt|_{\text{react}} \leq 0$; with Step 2, $d\mathcal{L}/dt \leq 0$.

Step 4 (LaSalle). Equality $d\mathcal{L}/dt = 0$ forces both contributions to vanish: $\xi \equiv 0$ a.e. (by positive definiteness) and $\nabla N = \nabla T = \nabla H = 0$ a.e. The largest invariant set contained in $\{d\mathcal{L}/dt = 0\}$ is thus the singleton $\{E_3\}$. The principle is applied along the orbit at hand, and $\mathcal{R}_*$ enters only through it: by Lemma S5.3, $u(\cdot, t) \in \mathcal{R}_*$ a.e. for $t \geq t_*$, a constraint closed in $\mathbf{H}$, so $\omega(u_0) \subseteq \overline{\{u(t) : t \geq t_*\}} \subseteq \mathcal{R}_*$, where $\mathcal{L}$ is finite and continuous. The three hypotheses are then in hand: the orbit $\{u(t) : t \geq t_*\}$ is precompact in $\mathbf{H}$ (Lemma 3.5 of the main text); $\mathcal{L}$ is continuous on $\mathcal{R}_*$ for the $\mathbf{H}$ -topology and non-increasing along the flow by Steps 1–3, the continuity resting on Lemma S5.3; and $\mathcal{L} \geq 0$. LaSalle's principle for infinite-dimensional semiflows, in the form asking for a precompact orbit and a continuous functional non-increasing along it [6, 7], then gives $\omega(u_0) \subseteq \{E_3\}$, that is $u(t) \rightarrow E_3$ in $L^2(\Omega)$. $\square$

Remark S5.5 (On the weights). Nothing forces $c_1 = c_2 = c_3 = 1$ in (S.8). Weights $c_i > 0$ produce the matrix $Q(c)$ with $Q_{11} = c_1$, $Q_{22} = c_2\beta$, $Q_{33} = c_3\omega$, $Q_{12} = c_2\alpha\beta/2$, $Q_{13} = c_1\delta/2$, $Q_{23} = -c_3\gamma/2$, and any c making $Q(c)$ positive definite yields the same conclusion, so optimising over c enlarges the certified region. The optimisation is explicit. Normalising by $c_1 = 1$ and working with $\det Q(c)/(c_1 c_2 c_3)$, scale-invariant where the determinant itself picks up λ^3 under $c \mapsto \lambda c$, one has

$$\frac{4 \det Q(c)}{c_2 c_3} = 4\beta\omega - \alpha\beta\delta\gamma - \left[\frac{c_3 \gamma^2}{c_2} + c_2 \alpha^2 \beta^2 \omega + \frac{\beta \delta^2}{c_3} \right], \quad (\text{S.10})$$

and the product of the three bracketed terms, $\alpha^2 \beta^3 \gamma^2 \delta^2 \omega$, does not depend on c . The arithmetic-geometric mean inequality bounds the bracket below by three times the cube root of that product, with equality at the unique point $c_2 = \rho^{2/3}/(\alpha^2 \beta)$, $c_3 = \delta^2/(\omega \rho^{2/3})$, $\rho := \alpha\delta/\delta_c$, whence the closed form

$$\max_c \frac{\det Q(c)}{c_2 c_3} = \beta\omega \left(1 - \frac{\rho}{4} - \frac{3}{4} \rho^{2/3} \right),$$

and with $r := \rho^{1/3}$ the identity $4 - r^3 - 3r^2 = -(r-1)(r+2)^2$ turns positivity into $r < 1$, that is, into $\rho < 1$, that is, into $q > 0$. Positive definiteness needs the whole Sylvester chain, and the first two minors come for free at c^{opt} : $m_1 = 1$ and $m_2 = c_2\beta(1 - \rho^{2/3}/4) > 0$ whenever $\rho < 1$, so the determinant is the binding condition and the equivalence above

holds for $Q(c^{\text{opt}}) \succ 0$. The Volterra family therefore certifies global stability exactly where E_3 is locally stable, and fails on branch (ii), where E_3 is a saddle and no such statement can hold: within this family the criterion is sharp. The computation, the stationarity of c^{opt} and the absence of any certificate when $\rho \geq 1$ are verified in Section 14 of the script described in Appendix A of the main text.

6 The threshold set of the spatial problem

This section proves the assertions of Remark 3.14 of the main text about the set

$$\Theta := \partial\mathcal{B}(E_1) \cap \mathcal{U}, \quad \mathcal{U} := \{ u_0 \in \mathcal{R} : N_0 \not\equiv 0 \text{ and } T_0 \not\equiv 0 \},$$

where $S(t)$ is the semiflow of (2.2) and $\mathcal{B}(E_i)$ collects the data $u_0 \in \mathcal{R}$ whose solution converges to E_i in $[L^\infty(\Omega)]^3$. Everything happens in $\mathbf{X}^\theta \hookrightarrow [C(\bar{\Omega})]^3$, where the cone K has non-empty interior, with neighbourhoods and boundaries relative to $\mathcal{R}$ in that topology.

Codimension of the stable set of E_3 . The global stable set, the union over n of the $S(n)$ -preimages of the local one, has the same codimension as the local one. Writing $W_{\text{loc}}^s = g^{-1}(0)$ for a submersion g onto $\mathbb{R}^4$ near E_3 , the preimage is $(g \circ S(n))^{-1}(0)$ and what needs checking is that $Dg \circ DS(n)$ is onto $\mathbb{R}^4$. It is, $DS(n)$ having dense range on $\mathbf{X}^\theta$ [6, §3.3, §6.1] and a continuous surjection onto a finite-dimensional space carrying any dense subspace onto the whole of it. Compactness of $DS(n)$ would not give this, its range being non-closed.

Strong monotonicity, precompactness, upward closure. The set $\mathcal{U}$ is relatively open, the two conditions reading $\|N_0\|_\infty > 0$ and $\|T_0\|_\infty > 0$ on non-negative data, and positively invariant by the strong maximum principle; it is the domain on which the semiflow is strongly monotone. Restricting to it costs nothing: a datum with $N_0 \equiv 0$ stays on the invariant face $\{N \equiv 0\}$, where T is logistic and H follows it, so the limit is E_2 ; symmetrically $T_0 \equiv 0$ gives E_1 , and both absent gives E_0 .

Precompactness in $\mathbf{X}^\theta$, which Lemma 3.5 of the main text gives in $\mathbf{H}$ alone, follows from parabolic smoothing. Solutions stay in $\mathcal{R}$, so $\|F(u(s))\|_{\mathbf{H}} \leq \sup_{\mathcal{R}} |F| \cdot |\Omega|^{1/2}$ along every orbit, and variation of constants from time $t - 1$, with the estimate

$$\|e^{\tau D \Delta_N} \varphi\|_{\mathbf{X}^{\theta'}} \leq C \tau^{-\theta'} \|\varphi\|_{\mathbf{H}}, \quad \theta' \in (\theta, 1), \quad \tau \in (0, 1],$$

of [6, Thm. 1.4.3], integrable at $\tau = 0$, bounds $\|u(t)\|_{\mathbf{X}^{\theta'}}$ for $t \geq 1$ by a constant made of $\sup_{\mathcal{R}} |F|$, $\sup_{v \in \mathcal{R}} |v|$ and $|\Omega|$, uniform in $t \geq 1$ and $u_0 \in \mathcal{R}$. The embedding $\mathbf{X}^{\theta'} \hookrightarrow \mathbf{X}^\theta$ being compact, $\{S(t)u_0 : t \geq 1, u_0 \in \mathcal{R}\}$ is precompact in $\mathbf{X}^\theta$. Two things follow: convergence in $[L^\infty(\Omega)]^3$ upgrades to convergence in $\mathbf{X}^\theta$; and $\mathcal{B}(E_1)$ is relatively open, being the union of the $S(t)$ -preimages of a neighbourhood of E_1 contained in it, which exists because the spectral bound of $J(E_1) - \mu D$ is negative and the linearised stability principle for sectorial operators [6, §5.1] applies.

Finally $\mathcal{B}(E_1)$ is closed upwards in $\mathcal{R}$: for $z \in \mathcal{B}(E_1)$ and $z \leq_K w \in \mathcal{R}$, E_1 being an equilibrium and the largest element of $\mathcal{R}$ for $\leq_K$, monotonicity gives $S(t)z \leq_K S(t)w \leq_K E_1$, hence $\|S(t)w - E_1\|_\infty \leq \|S(t)z - E_1\|_\infty \rightarrow 0$ and $w \in \mathcal{B}(E_1)$.

The two properties of Θ . They are standard in the theory of strongly monotone semiflows, for which we cite [8, 9, 10] in attribution: there they hold under hypotheses we have not checked, and the restricted forms we need are proved here.

Θ is invariant and contains no two K -comparable points. Invariance holds because $\mathcal{U}$ is positively invariant, because $S(t)x \in \mathcal{B}(E_1)$ would put x there too, and because $S(t)$ is continuous. Suppose then $x <_K y$ in Θ . Strong monotonicity on $\mathcal{U}$ gives $S(1)x \ll_K S(1)y$, both images lying in Θ . Since $S(1)x \in \partial\mathcal{B}(E_1)$ and $\text{int } K$ is open, some $z \in \mathcal{B}(E_1)$ is close enough to $S(1)x$ that $S(1)y - z$ still lies in $\text{int } K$; upward closedness then puts $S(1)y$ in $\mathcal{B}(E_1)$, contradicting $S(1)y \in \Theta$.

Θ is a Lipschitz graph of codimension one over any closed complement of a strictly positive direction of K . Take $e = (1, -1, -1)$, a constant in $\mathbf{X}^\theta$ and in the interior of K in $[C(\bar{\Omega})]^3$, so that $e + \varphi \in K$ for $\|\varphi\|_{\mathbf{X}^\theta} < r$ and some $r > 0$. Split $\mathbf{X}^\theta = V \oplus \mathbb{R}e$ with V closed and write $x = Px + \lambda(x)e$. For $x \neq y$ in Θ , $|\lambda(x - y)| > \|P(x - y)\|/r$ would make $x - y$ lie in K or in $-K$, which the previous paragraph forbids; hence $\|x - y\| \leq (1 + \|e\|/r) \|P(x - y)\|$, and Θ is the graph of a Lipschitz map from $P(\Theta) \subset V$ into $\mathbb{R}e$.

Irreducibility at a non-constant steady state. The instability of the non-constant states contained in Θ is read off the theorem of Kishimoto and Weinberger [11], which requires the linearisation at the state to be irreducible. The interior strong maximum principle gives $N \equiv 0$ or $N > 0$ on the open set Ω , and likewise for T . In the second case the couplings δN , $\alpha\beta T$ and γ are all non-zero and the linearisation is irreducible; in the first the state lies on an invariant face excluded from $\mathcal{B}$ by Lemma 3.16 of the main text, and no orbit issued from $\mathcal{B}$ reaches it.

We do not claim that Θ coincides with $\partial\mathcal{B}(E_2) \cap \mathcal{U}$ inside the order interval: that would need $\mathcal{B}(E_1) \cup \mathcal{B}(E_2)$ to be dense there, hence a third attractor with non-empty basin to be ruled out, and we have no proof of it.

7 Numerical controls for the size threshold

The critical radius quoted in Section 4.6 of the main text is obtained by bisection on the initial seed radius, the fate of a run being decided locally from the tumour peak and mass. Four controls, all performed by `critical_radius.py`, back the numbers reported there.

Refinement. On $\Omega = [0, 2]^2$ the bisection returns $R_c = 0.582 \pm 0.002$. Refining the mesh from $h = 0.02$ to $h = 0.0125$ moves it to 0.579, a shift of half a per cent and of the order of the bisection bracket itself; halving the time step leaves it at 0.582. The domain sweep uses $h = 0.02$ up to $L = 4$ and $h = 0.03$ beyond, the two agreeing at $L = 4$. In the ω sweep the mesh spacing is absolute where ℓ_H is not, so $\omega = 32$ is the least resolved point; halving h there moves R_c from 0.391 to 0.393 and R_c/ℓ_H from 2.21 to 2.22.

Margin of the invasion trigger. The invasion test fires when the tumour mass has grown by a quarter from its running minimum. On the two largest boxes, which carry the plateau $R_c^\infty = 2.31 \ell_H$, a near-critical focus that is eventually cleared rises to 1.10 and 1.07 times its running minimum, against a trigger at 1.25: the verdict is not a close call.

Monotonicity of the outcome in the seed radius. Bisection converges to a change of outcome and not necessarily to the first one, so the monotonicity it assumes is tested

rather than asserted. Fifteen radii spaced uniformly from 0.20 to 0.90 on $\Omega = [0, 2]^2$, and nine from 0.35 to 1.15 on the box carrying the plateau, give a single transition in each case, at 0.55–0.60 and at 0.75–0.85 respectively, which brackets the bisected values.

Saturation of $R_c(L)$ and the threshold state. The increment dR_c/dL falls from 0.30 on the first interval to 0.05 on $[3, 4]$ and to zero on $[4, 6]$: the last two bisections share a starting bracket and a number of halvings and return the same point of their common radius grid, so the two thresholds differ by less than 0.006. A saturating fit to the seven values gives an asymptote of 0.844, three per cent above the largest value measured, which is the accuracy to which the multiple 2.31 is fixed here; the main text quotes the measured value.

For the threshold state, the tumour mass retained while the near-critical trajectory lingers near the threshold is 0.235, 0.219, 0.195 and 0.161 for $L = 2, 2.5, 3$ and 3.5, against $T^* = 0.229$ at the coexistence state: within five per cent of it on the two smaller boxes and drifting below it on the two larger ones. The spatial standard deviation reached during the same phase, read on two runs per box, one on each side of the bisection bracket, is 0.15 and $0.33 T^*$ at $L = 2$, 0.40 and 0.64 at $L = 2.5$, 0.61 and 0.93 at $L = 3$, and 0.71 and 1.02 at $L = 3.5$.

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Author contributions (CRediT). **Yaya Youssouf Yaya:** Conceptualization, Methodology, Formal analysis, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing. **Barka Ibrahim Mahamat:** Methodology, Formal analysis, Validation, Investigation, Writing – review & editing. All authors have read and approved the final version of the manuscript.

Data availability. This manuscript reports a purely mathematical and computational analysis. No experimental, clinical, or otherwise externally acquired data are used. All numerical data displayed in the figures are reproducible from the self-contained Python code distributed with the submission, using the fixed pseudo-random seed declared in `params.py`.

Code availability. All code needed to reproduce every figure and every numerical claim in the paper is publicly available in the repository <https://github.com/yayayou47/acid-mediated-tumour-invasion>, archived under the persistent identifier 10.5281/zenodo.21878724 which resolves to the latest version of the archive and carries the nine figures as submitted. That repository is the reference copy: no code is bundled with this submission, so that a single version is cited, archived and maintained. It comprises the solver and figure

generators (`main.py`, `bifurcation_plane.py`, `bistability.py`, `fisher_kpp.py`), the bisection campaign behind the critical radius and the two panels of Figure 7 that report it (`critical_radius.py`, whose results are cached in `critical_radius_results.json`, distributed with the archive, and re-tabulated by `--report-only`), the single source of truth for the parameter values (`params.py`), and the symbolic and numerical verification of every algebraic claim made in the paper (`sympy_verification.py`). The random seed is fixed and no figure hard-codes a parameter, so the results are reproducible from a clean checkout.

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9 Symbolic verifications (SymPy)

The script `sympy_verification.py` accompanying this paper performs the symbolic computations underlying Theorem 3.9, Hypothesis (HL), Proposition 3.11 and the Sotomayor analysis of Section 3.6 of the main manuscript. We record here the three algebraic identities that are most easily overlooked in a hand calculation.

Routh–Hurwitz coefficients in E_3 . Expansion of $\det(J_3 - \lambda I)$ using the equilibrium relations yields the cubic $\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$ with $a_1 = N^* + \beta T^* + \omega$, $a_2 = N^*\beta T^* + N^*\omega + \beta T^*\omega$, $a_3 = \beta N^*T^*(\omega - \alpha\delta\gamma)$. The decomposition $a_1a_2 - a_3 = N^*\beta T^*(N^* + \beta T^*) + \omega(N^* + \beta T^*)^2 + \omega^2(N^* + \beta T^*) + \alpha\delta\gamma\beta N^*T^*$ is verified symbolically as an identity, all four summands being strictly positive on the existence region of E_3 .

Sylvester minors of Q at $c_i = 1$. $m_1 = 1$, $m_2 = \beta(4 - \alpha^2\beta)/4$, $m_3 = \det Q = \Delta/4$ with $\Delta = 4\beta\omega - \gamma^2 - \alpha^2\beta^2\omega - \alpha\beta\delta\gamma - \beta\delta^2$: positive-definiteness of Q is equivalent to Hypothesis (HL).

Sotomayor coefficients on Γ_δ . At $E_2 = (0, 1, \gamma/\omega)$ with $\delta = \delta_c$: $v = (-\omega/(\alpha\gamma), \omega/\gamma, 1)^\top$, $w = (1, 0, 0)^\top$, $w^\top F_\delta = 0$, $w^\top (DF)_\delta v = 1/\alpha$, $w^\top D^2F(v, v) = 2\omega^2(\alpha - 1)/(\alpha^2\gamma^2)$. The third coefficient vanishes precisely at $\alpha = 1$, flagging the codim-2 point P^* .

Segment of equilibria at P^* and its slow flow. Section 11 of the script checks that the reaction field vanishes identically on $\mathcal{S}$ at $(\alpha, \delta) = (1, \delta_c)$, that $J(s)\tau = 0$ for the tangent vector $\tau = (1, -1, -\gamma/\omega)^\top$ and $\ell(s)J(s) = 0$ for the left null vector $\ell(s) = (-\beta(1-s), s, \beta s(1-s)/\gamma)$, that the characteristic polynomial factors as $\lambda(\lambda^2 + A(s)\lambda + B(s))$ with the A, B quoted in the main text, that the restricted perturbed field equals $-s(1-s)(d, a\beta, 0)^\top$ exactly, and that the projection $(\ell F)/(\ell \tau)$ reproduces the reduced field (3.39) with zero residual.

Cooperative structure and global convergence. Section 13 of the script prints the off-diagonal entries of $\Sigma J \Sigma$ for $\Sigma = \text{diag}(1, -1, -1)$, namely δN , $\alpha\beta T$ and γ with three zeros, and the value of the field on each face of the open order interval, confirming that it points inwards on the four non-invariant faces. It then integrates the kinetics from eight random initial data for each of 120 random parameter vectors in each of the three non-bistable regions and reports the largest distance to the predicted limit: $1.4 \cdot 10^{-14}$, $5.3 \cdot 10^{-15}$ and $4.5 \cdot 10^{-16}$ respectively. Hypothesis (HL) fails on 83 of the 120 branch-(i) vectors without affecting the outcome, which is the numerical face of the fact that the order argument does not use it.

Modal count at the coexistence saddle. Section 12 evaluates the spectrum of $J(E_3) - \mu_{mn}D$ on the Neumann modes of $(0, 2)^2$ at the parameters of Section 4.5. Four modes carry a positive eigenvalue, namely $(0, 0)$, $(1, 0)$, $(0, 1)$ and $(1, 1)$, at $+0.161$, $+0.075$ twice and $+0.010$, and the largest eigenvalue crosses zero at $\mu^* = 5.3567$, which would require $L < 1.3574$ for all non-constant modes to be damped. Hence $\text{codim } W^s(E_3) = 4$ on the domain used, the count being unambiguous because $\mu^* \notin \{\mu_{mn}\}$ there, so that E_3 is hyperbolic for the spatial problem. The count applies to general perturbations. The centred initial data of Section 4.5 are invariant under the reflections of the square, and the three destabilising modes are odd under them, so those particular solutions evolve in a subspace where the first non-constant mode is $(2, 0)$, with $\mu_{20} = 9.87 > \mu^*$, and where $W^s(E_3)$ does separate the two basins.

Quasi-static acid profile of a disc source. Section 15 verifies the profile from which the critical seed radius of Section 4.5 is predicted. Writing $s = \sqrt{\omega}$, the field $H = (\gamma T_0/\omega)[1 - sRK_1(sR)I_0(sr)]$ inside the disc of radius R and $H = (\gamma T_0/\omega)sRI_1(sR)K_0(sr)$ outside solves $-\Delta H + \omega H = \gamma T_0 \mathbf{1}_{\{r < R\}}$ in radial coordinates with zero residual, and its derivative is continuous at $r = R$ identically. Continuity of H itself reduces to the Wronskian relation $I_0(z)K_1(z) + I_1(z)K_0(z) = 1/z$, which SymPy does not simplify on its own and which the script checks numerically to $7 \cdot 10^{-15}$. The value at the centre is the one quoted in the main text, and the bisection of $\delta H(0) = 1$ returns $R_c = 0.6396 = 1.809 \ell_H$ at the parameters of Section 4.5, with ω entering only through ℓ_H and the gauge-invariant ratio $\delta\gamma/\omega$.

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